
FINITE POSSIBILITY MECHANICS

MODULAR PAPER 05

Phenomenological Bridges

Route-Cost Correspondences with Quantum, Gravitational, and Cosmological
Observables

BRIDGE PAPER

Alx Spiker
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Abstract

Finite Possibility Mechanics supplies one microscopic accounting structure: a bounded directed route ledger, a constitutive viscosity field, a finite action budget, and local signed conservation. This paper develops the physical bridge layer that maps those internal quantities to established observables. The bridges cover pure dephasing, Landauer erasure, gravitational response, time dilation, cosmic acoustic structure, finite Born weights, Bell/CHSH correlations, and a bare electromagnetic coupling. The central result is architectural: the same route-cost variables enter every bridge, so the framework gains unification and accepts shared empirical risk.

The paper distinguishes four operations that are often conflated. An identity follows algebraically after definitions are fixed. A correspondence reproduces the form of an established equation. A calibration fixes dimensional scale from measured input. A prediction evaluates a pre-specified map on data not used to choose it. This classification keeps the five foundational axioms distinct from physical bridge laws that require an additional dimensional, dynamical, or observable dictionary.

Bridge result

FPM defines a coherent family of physical hypotheses rather than unrelated curve fits. The five foundational axioms determine the finite accounting structure; temperature, link maintenance, clock progress, and observable maps are separately declared bridge commitments. Tensor galaxy morphology, a complete cosmological likelihood, resource-dependent Bell correlations, and the observable meaning of the finite lag ceiling remain the decisive empirical frontier.

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1. Bridge methodology

1.1 Dependency stack

A physical bridge begins only after the foundational and executable layers are fixed. Paper 1 defines the local ledger and its conservation theorems. Paper 2 defines the real-valued reference runtime. Paper 3 defines the finite carrier, quantization, and information dynamics. Paper 4 defines an exact integer realization. The present paper asks which observable equations can be built from that shared machinery.

axioms $\rightarrow$ route ledger $R \rightarrow$ viscosity $\Omega \rightarrow$ action $L \rightarrow$ carrier $\Psi \rightarrow$ observable bridge

1.2 Evidence labels

- Derived: follows from stated definitions and earlier results without fitting an additional coefficient.
- Constitutive: a declared physical rule selecting one model from a wider admissible family.
- Calibrated: uses an observed dimensional scale or reference constant.
- Audited: evaluated numerically against an established equation or dataset.
- Predictive: fixed before application to data capable of rejecting it.

A bridge can occupy more than one category. For example, the dephasing recurrence is derived from the selected carrier update and then audited against a Lindblad channel. The gravitational constant calculation is algebraically determined after a temperature and scale map are selected, but its dimensional normalization is calibrated.

1.3 Dimensional discipline

Internal FPM variables are normalized. A bridge to SI observables therefore requires an explicit dimensional dictionary. Every such dictionary must identify its input units, calibration data, domain, and transformation law. Numerical proximity without that dictionary is not a bridge; a bridge is a reproducible map whose assumptions travel with its output.

2. Lindblad correspondence

For the two-state projection developed in Paper 3, the selected diagnostic leaves diagonal probabilities unchanged and multiplies the off-diagonal coherence by a persistence factor kappa. With the ordinary Hamiltonian-driven part held aside, the audited update is

$$\rho_{01,t+1} = \kappa \rho_{01,t}, \quad \rho_{00,t+1} = \rho_{00,t}, \quad \rho_{11,t+1} = \rho_{11,t}, \quad 0 \leq \kappa \leq 1.$$

For $\kappa = \exp(-\Gamma \Delta t)$, this is the finite-step pure-dephasing channel generated by the zero-Hamiltonian Lindblad form [5]. The executable audit uses the algebraically equivalent Euler parameterization $\gamma = (1 - \kappa) / \Delta t$. Positivity and trace preservation follow from the dephasing channel itself; no nonzero coherence target is introduced in this audit.

Status

This is an algebraic implementation correspondence for the selected zero-target pure-dephasing channel, not independent physical evidence. The physical hypothesis is that the persistence factor is controlled by FPM route burden rather than inserted as an independent environmental decay constant.

The runtime audit compares two implementations of that same recurrence and finds agreement down to machine precision. The discriminating experiment is not another reproduction of the selected curve; it is a test of whether measured decoherence changes with the internal route-cost variables predicted by FPM.

3. Landauer bridge and mass-equivalent scale

Consolidation removes representable alternatives from the finite carrier. The semantic entropy decrease is converted into a thermodynamic debit through Landauer's minimum erasure cost [6]

$$E_{\text{erase}} \geq k_B T \ln(2) \text{ per erased bit.}$$

The executable ledger records that debit subject to the energy actually available. Dividing an erasure energy by c^2 gives a mass-equivalent scale. Because the finite carrier contains N bit-equivalent slots, repeated erasure defines a discrete ladder rather than a continuous family:

$$m_n = n k_B T \ln(2) / c^2, \quad n = 0, 1, \dots, N.$$

This ladder is an energy-accounting result. A particle identification additionally requires stable dynamics, symmetry, charge, spin, lifetime, and interaction structure. The carrier capacity $N=1,452,997,909$ and its shell-filling radius are fixed by the finite construction in Paper 3, so the ladder spacing and endpoint are reproducible once the bridge temperature is specified.

Status

The information-to-energy inequality is established thermodynamics; applying it to FPM consolidation is a constitutive identification. The resulting mass-equivalent ladder is exact under that identification. A particle spectrum is a further dynamical problem.

3.1 Candidate nine-channel equipartition state law

The axioms use normalized route cost and do not select a Kelvin-valued temperature. A candidate dimensional extension introduces an energy-per-route dictionary ϵ_R and a thermodynamic state law. The directed 3 by 3 route ledger supplies $n_{\text{eff}}=9$ native channels. The additional equipartition postulate assigns the bulk node capacity E_{max} equally across those channels, so $\epsilon_R=h/[n_{\text{eff}} \Delta t_{\text{univ}}]$ and

$$T_{\text{erase}}=hE_{\text{max}}/[n_{\text{eff}}Nk_B \ln(2) \Delta t_{\text{univ}}]=306.442 \text{ K.}$$

A separate unweighted canonical-edge audit tests $g_{ij}=\ln[P_{ij}/P_{ji}]=-\beta_L(L_j-L_i)$ on the first master-chain action state after the tick-zero truth-target update. Across 375 undirected edges it gives $\text{corr}(\Delta L, g)=0.00469040864$, $\beta_L=-0.0019023970513$, and normalized residual $\|g+\beta_L \Delta L\|_2/\|g\|_2=0.999993565405$. The raw detailed-balance flux residual $\max|w_i P_{ij}-w_j P_{ji}|$ is 1.73×10^{-18} , while the separate log-ratio activity-potential residual $\max|g-\ln(w_j/w_i)|$ is 2.22×10^{-16} . That activity potential is not collinear with route-cost differences: the declared state has no scalar canonical temperature conjugate to L . This does not select a replacement state law.

The equipartition rule is a proposed constitutive closure, not a consequence of Axioms A1-A5 alone. Substitution into the present dimensional gravitational formula gives $G_{\text{FPM}}=6.53990 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$, 2.0137 percent below the CODATA value. The residual is an unresolved discrepancy, not an inferred screening correction. Any adoption must state the route-energy dictionary, the thermodynamic state law, and the conditions under which no scalar temperature is defined.

4. Emergent gravity and galaxy response

4.1 Route-cost gradients

FPM proposes that matter raises local route burden, route burden changes viscosity, and spatial viscosity gradients deflect propagation. In the continuum bridge, the effective acceleration is written as a baryonic source multiplied by a dimensionless susceptibility ν_{FPM} . The reference audit uses

$$\begin{aligned} v_{\text{FPM}}(x) &= 1 + [\Omega_{\text{max}}/\text{sqrt}(x+r_{\text{tensor}})] [1+E_Z x^2]^{-\beta}, \\ g_{\text{pred}} &= v_{\text{gb}}(x) g_{\text{bar}}, \quad v_{\text{gb}} = \max(10^{-9}, v_{\text{FPM}} - \alpha f_{\text{gas}}/(1+x)). \end{aligned}$$

Here $x=|g_{\text{bar}}|/a_0$ is the normalized baryonic-acceleration amplitude, r_{tensor} is the declared tensor-channel contribution, $E_Z=0.20E_{\text{max}}$ is the consolidation threshold, and β is inherited from the viscosity law.

The gas fraction is computed at each SPARC radius from the squared velocity contributions, $f_{\text{gas}} = \max(0, V_{\text{gas}} |V_{\text{gas}}|) / V_{\text{bar}}^2$, clipped to [0,1]. Thus f_{gas} is observed source composition, not a fitted exponent; $\alpha = 1/5$ is the trace-weight coefficient from Paper 1. The low-acceleration inverse-square-root response arises from the matter-load dilution law combined with the network's minimum-connectivity shift.

4.2 SPARC audit

The public reproducibility script downloads the SPARC master table and rotation-model archive from the cited Zenodo record [15] into a temporary directory, verifies their published MD5 checksums, and deletes them after the run. On its declared $Q=1$ sample of 99 galaxies with at least five valid rows, the gas-boundary FPM source functional gives median per-galaxy velocity RMSE 11.6065 km/s; the fixed comparison relation used by the script gives 11.7156 km/s. The comparison formula is defined by the archived implementation rather than by the 1983 modified-dynamics paper [7]. This is a reproducible retrospective scalar benchmark with no per-galaxy parameter fitting, not a tensor-morphology result, an independent confirmation, or a prospective prediction. The saved ledger records the source file checksums, selection rule, formulas, and metric.

Empirical obligation

The scalar rotation-curve audit must be followed by the harder tensor test: the full three-dimensional route ledger must predict non-axisymmetric morphology, lensing, and environmental dependence with one shared parameter ledger.

4.3 Dimensional gravitational scale

The implemented dimensional map is the following explicit chain. It uses the action ceiling through L_{max} , exact carrier capacity N , carrier radius α_{PP} derived by the shell fixed point in Paper 3 Section 6.1, Planck constant h , electron mass m_e , c , k_B , and the external bath calibration T_{bath} :

$$\begin{aligned} \zeta &= 9 / (4\pi L_{\text{max}}), \quad J = N k_B T_{\text{bath}} \ln 2, \\ \Delta t_{\text{univ}} &= h / (m_e c^2 \alpha_{\text{PP}}), \quad \Delta x_{\text{univ}} = c \Delta t_{\text{univ}}, \\ \mu_{\text{M,FPM}} &= (2/3) \zeta / [(\alpha_{\text{PP}} + 9) N^4], \\ G_{\text{FPM}} &= \mu_{\text{M,FPM}} \zeta c^4 \Delta x_{\text{univ}} / J. \end{aligned}$$

Equivalently, substitution gives $G_{\text{FPM}} = (2/3) \zeta^2 h c^3 / [m_e \alpha_{\text{PP}} (\alpha_{\text{PP}} + 9) N^5 k_B T_{\text{bath}} \ln 2]$. Under the candidate nine-channel equipartition state law, $T_{\text{bath}} = 306.442$ K and the map returns

$$G_{\text{FPM}} = 6.53990228 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}.$$

Compared with the CODATA value $6.67430(15) \times 10^{-11}$ in the same units [13], the relative difference is 2.0137 percent. The nine-channel count is native to the carrier; equal thermal partition is the added physical postulate. No unregistered factor is introduced to eliminate the remaining difference.

5. Internal lag and candidate muon-lag extension

Let L_{rest} be the per-tick action required to maintain a reference process and L the current route cost. The internal lag and its inverse causal throughput are

$$Y_{\text{FPM}} = L/L_{\text{rest}}, \quad X_{\text{throughput}} = L_{\text{rest}}/L.$$

The quantity $X_{\text{throughput}}$ decreases as route cost increases. It is an internal causal-throughput fraction, not a laboratory spatial velocity. The bounded action law gives

$$1 \leq Y_{\text{FPM}} \leq Y_{\text{max}} = 31.8738629.$$

5.1 Candidate muon-lag extension

A candidate motion-to-ledger dictionary reserves v_{lab} for an externally measured laboratory velocity and imports the ordinary low-load kinematic factor only as a dictionary:

$$L(v_{\text{lab}}) = \min[L_{\text{rest}} Y_{\text{kin}}(v_{\text{lab}}), L_{\text{max}}], \quad Y_{\text{FPM}}(v_{\text{lab}}) = \min[Y_{\text{kin}}, Y_{\text{max}}].$$

The candidate scheduler is $z_{t+1} = z_t + L_{\text{rest}}/L_t$. If muon decay is postulated to have a constant hazard per unit internal progress and the measured rest lifetime τ_{mu} is used only to set that hazard, the conditional survival law is

$$P_{\text{survive}}(t|v_{\text{lab}}) = \exp[-t/(\tau_{\text{mu}} Y_{\text{FPM}}(v_{\text{lab}}))].$$

This gives a ceiling at $v_* = 0.9995077253c$, $p_*/(mc) = 31.85817$, and $\tau_{\text{max}} = 70.0263$ microseconds for $\tau_{\text{mu}} = 2.1969811$ microseconds [16]. The supplied historical storage-ring reference gamma=29.33 lies below the conditional ceiling, so it does not distinguish this extension from the imported low-load kinematic map. The public feasibility package records no machine-readable controlled lifetime series above the threshold; a documented 10-50 GeV/c muon test-beam capability is not itself a decay-lifetime dataset.

The carrier phase update $\exp(-i \theta L)$ and the clock increment L_{rest}/L have opposite L -dependence. The runtime verifies their coexistence with unitary phase evolution and unchanged Born probabilities for a uniform phase rotation; that is an implementation consistency check, not evidence that either update represents a muon observable.

Status

The finite internal-lag ratio follows from the selected action bounds. The motion dictionary, decay-hazard identification, and rest-lifetime calibration are candidate extension commitments. Their capped high-gamma curve is a conditional competing prediction, not a result of Axioms A1-A5 alone.

6. Cosmology and acoustic structure

6.1 Common boot state

FPM replaces the need for distant regions to negotiate their initial uniformity with a common pre-execution state. A finite substrate can begin from one prepared condition, just as independently addressed storage blocks can share a format because they were initialized together. This is a proposed initial-condition mechanism, not superluminal communication between later regions.

6.2 Acoustic bridge

The numerical audit evaluates a compact algebraic spectrum rather than evolving a complete Boltzmann hierarchy. Its bridge function is

$$S(\ell) = A_{\text{FPM}} (\ell/\ell_A)^{n_s-1} \exp[-(\ell/\ell_D)^2] \{1 + A_{\text{osc}} \sin^2(\ell/\ell_{\text{osc}})\}.$$

The fixed quantities include $A_{\text{FPM}} = 4.0390 \times 10^{-5}$, $n_s = 0.9686263$, tensor ratio $r = 0.00348596$, damping scale $\ell_D = 1309.5688$, acoustic scale $\ell_A = 299.82$, and the declared ledger-inertia ratio 16/3. The compact template additionally sets $A_{\text{osc}} = 0.6$ and $\ell_{\text{osc}} = 80$. Those two oscillation parameters are fixed constitutive template inputs. The scalar local replenishment kernel is a dissipative first-order transfer rule and does not by itself generate acoustic phase oscillations. The displayed spectrum is therefore a phenomenological correspondence, not a derived acoustic-peak prediction; a physical CMB bridge requires additional transfer dynamics and a complete likelihood.

Against the selected Planck comparison [9], the existing fixed-nuisance audit reports Delta chi-squared = +4.16 relative to the Lambda-CDM reference. The positive value means the compact FPM bridge is somewhat worse in that comparison. Its importance is that a low-dimensional route-cost construction reaches the correct qualitative regime and exposes a complete next calculation: implement the transfer functions, polarization, lensing, covariance, foreground nuisance model, and a pre-committed likelihood protocol.

6.3 Tensor suppression

The bridge assigns one specific structural contraction of the nine-channel ledger to the tensor contribution, producing a one-in-nine suppression rule. This is a constitutive channel assignment, not a counting theorem: the 3 by 3 ledger has nine independent entries, while its trace is a contraction of three diagonal entries. A physical tensor derivation must connect that assignment to gauge-invariant perturbations and their propagation.

7. Finite Born distribution

For normalized carrier amplitudes ψ_a , FPM uses the Born-compatible weights $p_a = |\psi_a|^2$ squared [10] and realizes them with N finite microcells. Largest-remainder allocation returns integer counts n_a summing exactly to N , with each allocation error below one cell. For nine outcomes and the carrier capacity in Paper 3, the total-variation error is bounded by 3.10×10^{-9} ; the reported runtime error is 1.20×10^{-9} .

$$n_a/N \rightarrow |\psi_a|^2, \quad D_{\text{TV}} < m/(2N).$$

Pure phase rotation leaves the immediate weights unchanged. The current construction establishes finite realization in one selected channel basis. General detector orientation requires an explicit basis-change operation and contextual measurement map; that is the central extension needed before the bridge constitutes a general measurement theory.

8. Joint torsion and Bell/CHSH correlations

8.1 Explicit shared boundary

The FPM pair construction does not assign two independent local carriers prewritten answers. It quantizes a joint distribution across a shared torsion boundary. For detector settings a and b , the ideal correlation is

$$E(a,b)=-\cos[2(a-b)].$$

The four joint probabilities sum to one and leave each wing individually balanced at fifty-fifty. For the standard four settings, the CHSH combination reaches

$$S=|E(a,b)+E(a,b')+E(a',b)-E(a',b')|=2\sqrt{2}.$$

The runtime's finite allocation reproduces the target within its microcell error. An independent local baseline produces the expected triangular correlation and remains at or below $S=2$.

Locality classification

The shared torsion boundary is a non-Bell-local resource. The construction therefore does not evade Bell's theorem or the CHSH formulation [11,12]; it selects a joint architecture outside Bell factorizability and must satisfy no-signalling, relativistic causal consistency, and an independently specified formation and decay law.

8.2 Resource gate

$$q=1/\{1+\exp[10(\max(E_A,E_B)/(0.20E_{\max})-0.60)]\}.$$

Its sharpness, midpoint, and scale are constitutive inputs, and it describes joint-resolution activation at low stored energy rather than exhaustion of a paid link. The code's geometric correlation is $-\cos(a-b)$; the paper's $-\cos[2(a-b)]$ form requires an explicit setting convention, for example $\theta_{\text{code}}=2\theta_{\text{lab}}$.

8.3 Candidate torsion-refresh postulate

A resource-exhaustion mechanism requires an active maintenance operation even when the stored torsion configuration is pure gauge. A candidate postulate is that an experimentally usable shared link executes one refresh transaction per active independent antisymmetric generator per tick. For n_A generators its maintenance cost is $C_{\text{link}}=n_A c_0$. With a one-generator link, $c_0=0.05$, and a symmetric available action $A^{(0)}=0.5$ subject to the declared depletion law, the conditional snap threshold is

$$B_*=[A^{(0)}/(n_A c_0)]^{(4/3)}-1=20.5443469.$$

This threshold follows after the refresh postulate and availability map are adopted; it is not a result of the existing zero-cost pure-gauge link. The corresponding Bell transition must specify the laboratory control

variable, preparation distribution, and no-signalling constraints.

9. Bare electromagnetic coupling

The torsion-snap bridge uses two distinct floors that must not be conflated. The action floor is $c_0=0.05$. The quantity $e_{\text{floor}}=0.0314$ is instead the dimensionless structural percolation floor in the causal-depletion law

$$e_{\text{eff}}(B)=\max[(1+B)^{-3/4}, e_{\text{floor}}].$$

Here $B \geq 0$ is the dimensionless local baryonic, or ordinary-matter, load supplied to the viscosity-energy gate; $B=0$ denotes no added matter load. It is a bridge input distinct from cache bias b , bit count B_{erase} , and the action L .

Thus e_{floor} is not c_0/E_{max} , which equals 0.075. With the symmetric-sector exponent $\beta=9/5$, the maximum symmetric occupancy is

$$C_{\text{sym,max}}=(1/e_{\text{floor}})^{1/\beta}, \quad \alpha_{\text{bare}}=c_0/C_{\text{sym,max}}.$$

Using the shared FPM constants gives inverse $\alpha_{\text{bare}} = 136.7946397586$. The CODATA inverse fine-structure constant is approximately 137.036 [13]; the difference is about 0.1761 percent. FPM interprets its number as a bare coupling before vacuum screening, not as the macroscopic measured value itself. The physical programme is therefore to derive the screening correction from carrier excitations rather than choosing a correction after comparison.

The proposed excitation is a paid, trace-free transverse disturbance released when a shared topological link can no longer be maintained. The decomposition of the 3 by 3 ledger into antisymmetric and symmetric sectors supplies three circulation-like and six strain-like components. Connecting this structure to a massless spin-one gauge field, charge conservation, polarization, and quantum electrodynamic running remains the decisive field-theory derivation.

Status

The bare number is an exact output of the selected upstream constants. Establishing independence from the target requires sensitivity analysis and a derivation whose choices are fixed before comparison with α .

10. Candidate extension programs and empirical risk

The bridges are coupled through route cost. This is the framework's central economy and its strongest vulnerability. A failure of the local ledger or viscosity law propagates into gravity, time dilation, cosmology, and the coupling construction; a failure unique to a detector map need not invalidate local conservation or the finite carrier.

- Ledger layer: locality, signed conservation, and finite support are shared by every bridge.
- Constitutive layer: viscosity endpoints and exponents jointly affect gravity, lag, and several reported constants.

- Carrier layer: normalization, phase, consolidation, and finite allocation jointly affect Born, Bell, and Landauer results.
- Nine-channel equipartition extension: an energy-to-route dictionary and equal thermal partition across the nine directed channels determine a Kelvin-valued temperature and propagate directly into G and mass-equivalent scales.
- Torsion extension: a refresh postulate and a control-variable map determine whether a shared link has a resource-dependent survival threshold.
- Lag extension: a progress scheduler and clock dictionary determine whether `gamma_FPM` maps to a measured rate or lifetime.
- Observable layer: SPARC source functions, cosmological transfer functions, detector bases, and screening maps can fail independently.

10.1 Highest-value next tests

- Predict full two-dimensional galaxy velocity fields and lensing from the tensor ledger, with one frozen parameter set.
- Construct a dynamical cosmological transfer model before treating acoustic structure as a physical prediction.
- Adopt and test a progress scheduler plus a concrete clock or particle observable for `gamma_FPM`.
- Adopt a torsion-refresh postulate, specify its laboratory resource variable, and test the predicted CHSH transition while preserving no-signalling.
- Derive electromagnetic screening and scale dependence from the same carrier rather than importing the measured correction.

11. Conclusion

FPM's bridge programme is an auditable dependency stack. The foundational ledger supplies local transport, signed accounting, finite allocation, and fixed action bounds. Temperature, physical link maintenance, clock progress, and cosmological transfer dynamics are candidate extension programs with explicit new commitments. Calibrated and correspondence-level numerical comparisons remain distinct from physical predictions.

Central conclusion

A physical bridge is valid only at the level warranted by its declared dependencies. The extension programs make their additional assumptions, derived conditional consequences, and failure conditions explicit.

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